

CE/CZ4055: CPS Security
Tutorial 3

- (1) Explain the differences between digital signatures and MACs with respect to the property of Non-Repudiation. Non-Repudiation is a crucial property required for legal purposes. So can Digital signatures be directly used for legal purposes? If not, explain what can go wrong if one uses Digital signatures for legal purposes.
- (2) Suppose Alice and Bob use "TextBook RSA" to communicate. Let the two primes chosen by Alice is $p = 17$ and $q = 23$. Demonstrate with suitable simple examples, key generation, encryption and decryption for the given parameters.
- (3) Alice and Bob are again using "TextBook RSA" to communicate. They are using the same modulus n , so that Alice's public encryption key is (n, e_A) and Bob's is (n, e_B) . Alice and Bob happen to have chosen e_A and e_B that are relatively prime. Now Charles wants to send the message m to both Alice and Bob. You may assume that $m < n$.
 - (a) What does Charles send to Alice? What does he send to Bob?
 - (b) Suppose that Eve intercepts both of these transmissions. Show how Eve can recover m without factoring n .

q3. Alice (n, e_A) , bob (n, e_B) . $C_a = m^{e_A}$, $C_b = m^{e_B}$. Bezout: Identity x and y are prime, is mathematically proven that there exists a S and T where $Sx + Ty = 1$. Eve will be able to find s and t . Eve can use $C_a^s * C_b^t = (m^{e_A} \bmod n)^s * (m^{e_B} \bmod n)^t = (m^{e_A*s} * m^{e_B*t}) \bmod n = m^{(e_A*s + e_B*t)} \bmod n = m' \bmod n = m$. Thus e_A and e_B should not be prime.